

Izuma Device Management

Chip-to-Cloud IoT at Scale

Building custom IoT device management infrastructure takes months to years. Izuma Device Management compresses that timeline to days — a proven platform with over a decade of evolution (originally Arm Pelion / mbed Cloud) handling secure connectivity, firmware updates, and fleet operations from ultra-constrained MCUs to edge servers. **SOC 2 certified.**

Core Capabilities

Secure Device Connectivity

LwM2M/CoAP with DTLS. X.509 mTLS. Hardware root of trust (TPM 2.0, TrustZone, PARSEC). Protocol translation for legacy devices.

Firmware Updates (FOTA)

One-shot, continuous, mesh campaigns. Delta updates. Encrypted and signed images. Resume after disconnect.

Fleet Management

Directory scales from dozens to millions. Filter, group, suspend, bulk-operate. Custom attributes.

Edge Computing & Monitoring

True Kubernetes. Offline containers. SD-WAN. Device Sentry threshold-based incidents. Remote terminal.

Device Spectrum

Device Class	Stack
Constrained MCUs	Izuma Connect (Lite)
Mainstream MCUs / MPUs	Izuma Connect
Rich nodes (Linux)	Izuma Connect
Gateways	Izuma Edge
Edge servers (k8s)	Izuma Edge

Deployment

Model	Infrastructure
Fully Managed	SaaS on AWS (NoOps)
Any Cloud	AWS, Azure, GKE
Bare Metal	On-premises

Why Not Build It Yourself?

Consideration	Build In-House	Izuma Device Management
Time to production	Months to years	Days
Security architecture	Design from scratch	Chip-to-cloud, SOC 2 certified
Protocol support	Implement each protocol	LwM2M, CoAP, BLE, Zigbee, Modbus built in
Firmware & edge	Build pipeline + k8s integration	Delta/mesh campaigns, native k8s, SD-WAN
Factory provisioning	Custom tooling required	Factory Flow — works offline
Ongoing maintenance	Permanent engineering team	Managed service option (NoOps)